

Waterfowl

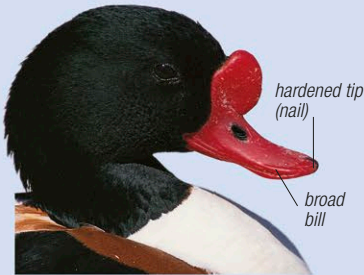
PHYLUM	Chordata
CLASS	Aves
ORDER	Anseriformes
FAMILIES	3
SPECIES	177

Also known as wildfowl, members of this group include ducks, geese, and swans, as well as a family of 3 South American species known as screamers. Aided by waterproof plumage and webbed feet, they are among the

dominant birds of freshwater wetlands. They are also found in estuaries and inshore coastal waters, while a few species are entirely marine. Waterfowl are strong swimmers. They feed mostly from the surface of water, although many ducks dive in search of food, while some species (notably geese, swans, and screamers) graze on land. Waterfowl are powerful fliers, and some species undertake annual migrations of thousands of miles between their wintering and breeding areas.

Anatomy

Waterfowl have a plump, buoyant body, with a small head and, usually, a short tail. Most species have a broad, flattened bill (see below) and a long neck for reaching down to feed underwater. Many species are gaudily colored. When molting, most lose all their flight feathers simultaneously. To hide them from predators during the period when they are unable to fly, the males (drakes) acquire a drab coloring, known as an eclipse plumage. Waterfowl are insulated from cold by a layer of down and a layer of fat beneath the skin. Screamers look different from other waterfowl in several respects: they have longer legs, spurred wings, a chickenlike bill, and toes webbed only at the base.



BILL
Like most waterfowl, the common shelduck has a wide, flattened bill. The edges have small ridges, called lamellae, which are used to grip prey or filter edible particles out of the water. The tip of the upper bill may be hardened into a nail for tearing vegetation.

PREENING

Plumage care is important for waterfowl such as this greylag goose, because it is essential that their feathers remain waterproof. A gland on the bird's rump secretes a water-repellent oil. The bird stimulates the gland with its bill and then spreads the oil around its body by rubbing and preening its feathers.



FEET
Waterfowl use their webbed feet to help propel themselves through water. Only the front 3 toes are webbed; the smaller hind toe is raised. This enables waterfowl to walk on land, although they do so with a waddling gait.

Reproduction

The reproductive cycle of waterfowl is suited to a life spent on water, which often involves exposure to cold and predators. Unusual among birds, the males have an organ similar to a penis, which is inserted inside the female's cloaca, making it possible for them to mate on the surface of water. Waterfowl usually nest near water, most often among ground vegetation, although some species use holes in trees or rock crevices. The females of some species, such as eiders, pluck their own down to make an insulating cover for their eggs.



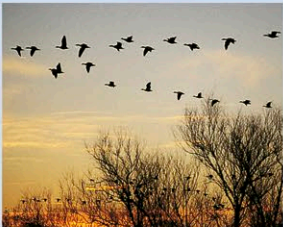
YOUNG
Waterfowl hatch in a well-developed state, with their eyes open and a covering of down that dries quickly. Hours after hatching, the chicks (such as these Canada geese) can walk or swim. Leaving the nest, they follow the adult to the relative safety of water, and soon begin to search for their own food.



TAKING OFF
Although some waterfowl can spring almost vertically from water, the heavier species have to resort to a running takeoff. Swans, in particular, need a long runway, gaining speed by pattering with their feet across the surface of the water.

Movement

Several adaptations help waterfowl move easily through water. The smooth outline of their body reduces water resistance, while their webbed feet (see left) act as powerful paddles. Thick plumage adds to their buoyancy, although many duck species that dive to find food sleek their plumage to expel air and reduce their buoyancy before submerging. With their huge wings and exceptionally light bones, screamers can fly for hours on end, often soaring. Other species of waterfowl have smaller wings that must be flapped constantly when in flight. Once airborne, however, they are fast and powerful fliers, with some species capable of exceeding 60mph (100kph) in level flight.



FLYING FORMATION
Flocks of waterfowl (here, snow geese) fly in a characteristic V-formation. By staying in the slipstream of the leading bird, those further back meet with less turbulence and thereby save energy. The position of hard-working leader is changed regularly.